Integration

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Volumes of revolution: the disk method**

The region under $y = \sqrt{x}$ on $[0, 4]$ is revolved about the x-axis. Find the volume generated, using $V = \pi\int y^2\,dx$.

$V = \pi\int_0^4 x\,dx = \pi\left[\frac{x^2}{2}\right]_0^4 = 8\pi$.

2 **[Foundational]** • **Target: Integration by substitution**

Find $\int 2x(x^2+1)^4\,dx$ using the substitution $u = x^2 + 1$.

$du = 2x\,dx$. Integral becomes $\int u^4\,du = \frac{u^5}{5} + C = \frac{(x^2+1)^5}{5} + C$.

3 **[Foundational]** • **Target: Integration by substitution**

Find $\int x\sqrt{x^2+9}\,dx$ using the substitution $u = x^2 + 9$.

$du = 2x\,dx$, so $x\,dx = \frac{du}{2}$. Integral becomes $\int \sqrt{u}\cdot\frac{du}{2} = \frac{1}{2}\cdot\frac{2}{3}u^{3/2} + C = \frac{1}{3}(x^2+9)^{3/2} + C$.

4 **[Foundational]** • **Target: Integration by substitution**

Find $\int (2x+1)(x^2+x)^3\,dx$ using the substitution $u = x^2 + x$.

$du = (2x+1)\,dx$. Integral becomes $\int u^3\,du = \frac{u^4}{4} + C = \frac{(x^2+x)^4}{4} + C$.

5 **[Foundational]** • **Target: Integration by substitution**

Find $\int 3x^2(x^3-2)^4\,dx$ using the substitution $u = x^3 - 2$.

$du = 3x^2\,dx$. Integral becomes $\int u^4\,du = \frac{u^5}{5} + C = \frac{(x^3-2)^5}{5} + C$.

6 **[Foundational]** • **Target: Integration by parts**

Find $\int xe^x\,dx$ using integration by parts.

Let $u = x$, $dv = e^x\,dx$, so $du = dx$, $v = e^x$. $\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C = e^x(x-1) + C$.

7 **[Foundational]** • **Target: Integration by parts**

Find $\int x\cos x\,dx$ using integration by parts.

Let $u = x$, $dv = \cos x\,dx$, so $du = dx$, $v = \sin x$. $\int x\cos x\,dx = x\sin x - \int \sin x\,dx = x\sin x + \cos x + C$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 **[Intermediate]** • **Target: Integration by parts**

Find $\int \ln x \, dx$ using integration by parts with $u = \ln x$ and $dv = dx$.

$du = \frac{1}{x}dx$, $v = x$. $\int \ln x \, dx = x\ln x - \int x \cdot \frac{1}{x} dx = x\ln x - \int 1 \, dx = x\ln x - x + C$.

9 **[Intermediate]** • **Target: Integration by parts**

Find $\int x^2 e^x \, dx$ using integration by parts twice.

Let $u = x^2$, $dv = e^x \, dx$: $\int x^2 e^x \, dx = x^2 e^x - \int 2xe^x \, dx$. From an earlier result, $\int xe^x \, dx = e^x(x-1) + C$, so $\int 2xe^x \, dx = 2e^x(x-1) + C$. Result: $x^2e^x - 2e^x(x-1) + C = e^x(x^2 - 2x + 2) + C$.

10 **[Intermediate]** • **Target: Integration using partial fractions**

Find $\int \frac{5x+7}{(x+1)(x+2)} dx$, first decomposing into partial fractions.

$\frac{5x+7}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$: $5x + 7 = A(x+2) + B(x+1)$. At $x = -1$: $A = 2$. At $x = -2$: $B = 3$. Integral: $2\ln|x+1| + 3\ln|x+2| + C$.

11 **[Intermediate]** • **Target: Integration using partial fractions**

Find $\int \frac{1}{x^2 - 4} dx$ using partial fractions, factoring $x^2 - 4 = (x-2)(x+2)$.

$\frac{1}{(x-2)(x+2)} = \frac{A}{x-2} + \frac{B}{x+2}$: $1 = A(x+2) + B(x-2)$. At $x = 2$: $A = \frac{1}{4}$. At $x = -2$: $B = -\frac{1}{4}$. Integral: $\frac{1}{4}\ln\left|\frac{x-2}{x+2}\right| + C$.

12 **[Intermediate]** • **Target: Integration using partial fractions**

Find $\int \frac{3x-1}{x(x-3)} dx$, first decomposing into partial fractions.

$\frac{3x-1}{x(x-3)} = \frac{A}{x} + \frac{B}{x-3}$: $3x - 1 = A(x-3) + Bx$. At $x = 0$: $A = \frac{1}{3}$. At $x = 3$: $B = \frac{8}{3}$. Integral: $\frac{1}{3}\ln|x| + \frac{8}{3}\ln|x-3| + C$.

13 **[Intermediate]** • **Target: Integration using partial fractions**

Find $\int \frac{2x+3}{(x-1)(x+4)} dx$, first decomposing into partial fractions.

$\frac{2x+3}{(x-1)(x+4)} = \frac{A}{x-1} + \frac{B}{x+4}$: $2x + 3 = A(x+4) + B(x-1)$. At $x = 1$: $A = 1$. At $x = -4$: $B = 1$. Integral: $\ln|x-1| + \ln|x+4| + C$.

14 **[Intermediate]** • **Target: Integrating trigonometric, exponential and logarithmic functions**

Find $\int \sin(3x) \, dx$ and $\int e^{2x} \, dx$.

$\int \sin(3x) \, dx = -\frac{1}{3}\cos(3x) + C$. $\int e^{2x} \, dx = \frac{1}{2}e^{2x} + C$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Integrating trigonometric, exponential and logarithmic functions • Cross-Lesson: 6.01 Sequences & Series

Find $\int \frac{4x}{x^2+1}\,dx$.

Since $\frac{d}{dx}[\ln(x^2+1)] = \frac{2x}{x^2+1}$, and $\frac{4x}{x^2+1} = 2 \cdot \frac{2x}{x^2+1}$: $\int \frac{4x}{x^2+1}\,dx = 2\ln(x^2+1) + C$.

16 [Challenging] • Target: Integrating trigonometric, exponential and logarithmic functions • Cross-Lesson: 7.01 Vectors

Find $\int \tan x\,dx$, using the substitution $u = \cos x$.

$\tan x = \frac{\sin x}{\cos x}$. With $u = \cos x$, $du = -\sin x\,dx$: $\int \frac{\sin x}{\cos x}\,dx = -\int \frac{du}{u} = -\ln|u| + C = -\ln|\cos x| + C = \ln|\sec x| + C$.

17 [Challenging] • Target: Integrating trigonometric, exponential and logarithmic functions • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Find $\int \frac{1}{2x+3}\,dx$.

$\int \frac{1}{2x+3}\,dx = \frac{1}{2}\ln|2x+3| + C$.

18 [Challenging] • Target: Definite integrals using substitution • Cross-Lesson: 9.01 Complex Numbers

Evaluate $\int_0^2 x(x^2+1)^3\,dx$ using the substitution $u = x^2+1$.

When $x = 0$, $u = 1$; when $x = 2$, $u = 5$; $x\,dx = \frac{du}{2}$. $\int_1^5 u^3 \cdot \frac{du}{2} = \frac{1}{2}\left[\frac{u^4}{4}\right]_1^5 = \frac{1}{8}(625-1) = \frac{624}{8} = 78$.

19 [Challenging] • Target: Definite integrals using substitution • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Evaluate $\int_0^{\pi/2} \sin x\cos^3 x\,dx$ using the substitution $u = \cos x$.

$du = -\sin x\,dx$. When $x = 0$, $u = 1$; when $x = \frac{\pi}{2}$, $u = 0$. $\int_1^0 u^3(-du) = \int_0^1 u^3\,du = \left[\frac{u^4}{4}\right]_0^1 = \frac{1}{4}$.

20 [Challenging] • Target: Definite integrals using substitution • Cross-Lesson: 10.01 Numerical Methods

Evaluate $\int_1^e \frac{\ln x}{x}\,dx$ using the substitution $u = \ln x$.

$du = \frac{1}{x}dx$. When $x = 1$, $u = 0$; when $x = e$, $u = 1$. $\int_0^1 u\,du = \left[\frac{u^2}{2}\right]_0^1 = \frac{1}{2}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Standard Integrals (Trig, Exp & Rational)

Find $\int \frac{2}{3x-1}\,dx$.

a. $\frac{2}{3}\ln|3x-1| + C$ ✓

b. $2\ln|3x-1| + C$

c. $\frac{1}{3}\ln|3x-1| + C$

d. $-\frac{2}{(3x-1)^2} + C$

$\int \frac{1}{ax+b}dx = \frac{1}{a}\ln|ax+b| + C$. Multiplying by 2 gives $\frac{2}{3}\ln|3x-1| + C$.

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22 [Foundational] • Target: Standard Integrals (Trig, Exp & Rational)

Find $\int \cos^2(x)\,dx$.

a. $\frac{x}{2} + \frac{\sin(2x)}{4} + C$ ✓

b. $\frac{\sin^3(x)}{3} + C$

c. $\frac{x}{2} - \frac{\sin(2x)}{4} + C$

d. $\sin(2x) + C$

Use identity $\cos^2 x = \frac{1+\cos(2x)}{2}$. $\int \frac{1+\cos(2x)}{2}dx = \frac{x}{2} + \frac{\sin(2x)}{4} + C$.

23 [Foundational] • Target: Standard Integrals (Trig, Exp & Rational)

Evaluate $\int_0^{\pi/4} \tan(x)\,dx$.

a. $\frac{1}{2}\ln 2$ ✓

b. $\ln 2$

c. 1

d. $\frac{\sqrt{2}}{2}$

$\int \tan x dx = [-\ln|\cos x|]_0^{\pi/4} = -\ln(\sqrt{2}/2) + 0 = -\ln(2^{-1/2}) = \frac{1}{2}\ln 2$.

24 [Foundational] • Target: Integration by Substitution

Evaluate $\int x\sqrt{x^2+5}\,dx$.

a. $\frac{1}{3}(x^2+5)^{3/2} + C$ ✓

b. $\frac{1}{2}(x^2+5)^{3/2} + C$

c. $\frac{2}{3}(x^2+5)^{3/2} + C$

d. $(x^2+5)^{3/2} + C$

Let $u = x^2+5$, $du = 2xdx$. $\int \frac{1}{2}u^{1/2}du = \frac{1}{2}\cdot\frac{2}{3}u^{3/2} + C = \frac{1}{3}(x^2+5)^{3/2} + C$.

25 [Foundational] • Target: Integration by Substitution

Evaluate $\int_0^1 \frac{x}{x^2+1}\,dx$.

a. $\frac{1}{2}\ln 2$ ✓

b. $\ln 2$

c. $\frac{\pi}{4}$

d. 1

Let $u = x^2+1$, $du = 2xdx$. Integral from $u=1$ to $u=2$: $\int_1^2 \frac{1}{2u}du = [\frac{1}{2}\ln u]_1^2 = \frac{1}{2}\ln 2$.

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26 [Foundational] • Target: Integration by Substitution

Use the substitution $x = 2\sin\theta$ to find $\int \sqrt{4 - x^2}\, dx$.

a. $2\theta + \sin(2\theta) + C$ ✓

b. $4\theta + 2\sin(2\theta) + C$

c. $2\arcsin(x/2) + C$

d. $\frac{1}{2}(4 - x^2)^{3/2} + C$

$dx = 2\cos\theta d\theta$, $\sqrt{4 - x^2} = 2\cos\theta$. $\int 4\cos^2\theta d\theta = \int 2(1 + \cos 2\theta)d\theta = 2\theta + \sin(2\theta) + C$.

27 [Foundational] • Target: Integration by Parts

Find $\int xe^{3x}\, dx$.

a. $\frac{xe^{3x}}{3} - \frac{e^{3x}}{9} + C$ ✓

b. $\frac{xe^{3x}}{3} - \frac{e^{3x}}{3} + C$

c. $xe^{3x} - e^{3x} + C$

d. $\frac{x^2e^{3x}}{6} + C$

Let $u = x, dv = e^{3x}dx \Rightarrow du = dx, v = \frac{1}{3}e^{3x}$. $uv - \int v du = \frac{1}{3}xe^{3x} - \int \frac{1}{3}e^{3x}dx = \frac{xe^{3x}}{3} - \frac{e^{3x}}{9} + C$.

28 [Foundational] • Target: Integration by Parts

Evaluate $\int_1^e x\ln x\, dx$.

a. $\frac{e^2 + 1}{4}$ ✓

b. $\frac{e^2 - 1}{4}$

c. $\frac{e^2}{2}$

d. $\frac{e^2 - 1}{2}$

$u = \ln x, dv = xdx \Rightarrow du = dx/x, v = x^2/2$. $[\frac{x^2}{2}\ln x]_1^e - \int_1^e \frac{x}{2}dx = \frac{e^2}{2} - [\frac{x^2}{4}]_1^e = \frac{e^2}{2} - (\frac{e^2}{4} - \frac{1}{4}) = \frac{e^2+1}{4}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Integration by Parts

Evaluate $\int_0^\pi x\sin x\, dx$.

a. π ✓

b. 2π

c. 0

d. 2

$u = x, dv = \sin x dx \Rightarrow du = dx, v = -\cos x$. $[-x\cos x]_0^\pi - \int_0^\pi (-\cos x)dx = -\pi(-1) - 0 + [\sin x]_0^\pi = \pi + 0 = \pi$.

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30 [Intermediate] • Target: Partial Fractions Integration & Trapezium Rule

Find $\int \frac{5x-2}{(x-2)(x+1)}\,dx$.

a. $\frac{8}{3}\ln|x-2|+\frac{7}{3}\ln|x+1|+C$ ✓

b. $3\ln|x-2|+2\ln|x+1|+C$

c. $8\ln|x-2|+7\ln|x+1|+C$

d. $\ln|(x-2)(x+1)|+C$

Partial fractions: $\frac{A}{x-2}+\frac{B}{x+1}$. For A: $(5(2)-2)/(2+1)=8/3$. For B: $(5(-1)-2)/(-1-2)=-7/(-3)=7/3$. Integrating gives $\frac{8}{3}\ln|x-2|+\frac{7}{3}\ln|x+1|+C$.

31 [Intermediate] • Target: Partial Fractions Integration & Trapezium Rule

Use the trapezium rule with 4 intervals (5 ordinates) to approximate $\int_1^3 \frac{1}{x}\,dx$.

a. 1.117 ✓

b. 1.099

c. 1.150

d. 1.050

$h=(3-1)/4=0.5$. $x=1, 1.5, 2, 2.5, 3$. $y=1, 2/3, 1/2, 2/5, 1/3$. Area $\approx \frac{0.5}{2}[1+1/3+2(2/3+1/2+2/5)] = 0.25[4/3+2(47/30)] = 0.25[40/30+94/30] = 0.25(134/30) = 134/120 \approx 1.1167$.

32 [Intermediate] • Target: Partial Fractions Integration & Trapezium Rule

Is the trapezium rule approximation in the previous question an overestimate or underestimate, and why?

a. **Overestimate, because the curve $y=1/x$ is concave up** ✓

b. Underestimate, because $y=1/x$ is decreasing

c. Overestimate, because the intervals are too wide

d. Underestimate, because the chords lie below the curve

Since $d^2y/dx^2 = 2/x^3 > 0$ for $x>0$, the curve is convex (concave up). The straight line chords lie above the curve, producing an overestimate.

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33 [Intermediate] • Target: Integration by substitution

Using the trapezium rule with 5 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 15, y_1 = 13, y_2 = 3, y_3 = 15, y_4 = 10, y_5 = 6$.

a. $\frac{619}{12}$

b. $\frac{105}{2}$

c. 31

d. $\frac{103}{2}$ ✓

$\int y\,dx \approx \frac{h}{2}[y_0 + y_5 + 2(y_1 + \cdots + y_4)] = \frac{1}{2} \times 103 = \frac{103}{2}$.

34 [Intermediate] • Target: Integration by parts

Using the trapezium rule with 4 strips of width $h = 3$, estimate $\int y\,dx$ given $y_0 = 8, y_1 = 18, y_2 = 5, y_3 = 18, y_4 = 17$.

a. $\frac{1927}{12}$

b. $\frac{321}{2}$ ✓

c. 99

d. $\frac{327}{2}$

$\int y\,dx \approx \frac{h}{2}[y_0 + y_4 + 2(y_1 + \cdots + y_3)] = \frac{3}{2} \times 107 = \frac{321}{2}$.

35 [Intermediate] • Target: Integration using partial fractions

Using the trapezium rule with 6 strips of width $h = 3$, estimate $\int y\,dx$ given $y_0 = 4, y_1 = 3, y_2 = 19, y_3 = 3, y_4 = 16, y_5 = 2, y_6 = 17$.

a. $\frac{1927}{12}$

b. $\frac{321}{2}$ ✓

c. 96

d. $\frac{327}{2}$

$\int y\,dx \approx \frac{h}{2}[y_0 + y_6 + 2(y_1 + \cdots + y_5)] = \frac{3}{2} \times 107 = \frac{321}{2}$.

36 [Intermediate] • Target: Integrating trigonometric, exponential and logarithmic functions

Using the trapezium rule with 6 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 9, y_1 = 1, y_2 = 12, y_3 = 20, y_4 = 11, y_5 = 17, y_6 = 12$.

a. $\frac{145}{2}$

b. $\frac{859}{12}$

c. 41

d. $\frac{143}{2}$ ✓

$\int y\,dx \approx \frac{h}{2}[y_0 + y_6 + 2(y_1 + \cdots + y_5)] = \frac{1}{2} \times 143 = \frac{143}{2}$.

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37 [Intermediate] • Target: Definite integrals using substitution

Using the trapezium rule with 4 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 16, y_1 = 15, y_2 = 14, y_3 = 19, y_4 = 7$.

a. $\frac{715}{12}$

b. $\frac{71}{2}$

c. $\frac{121}{2}$

d. $\frac{119}{2}$ ✓

$\int y\,dx \approx \frac{h}{2}[y_0 + y_4 + 2(y_1 + \cdots + y_3)] = \frac{1}{2} \times 119 = \frac{119}{2}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Volumes of revolution: the disk method • Cross-Lesson: 9.01 Complex Numbers

Using the trapezium rule with 3 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 1, y_1 = 1, y_2 = 2, y_3 = 10$.

a. $\frac{17}{2}$ ✓

b. 7

c. $\frac{103}{12}$

d. $\frac{19}{2}$

$\int y\,dx \approx \frac{h}{2}[y_0 + y_3 + 2(y_1 + \cdots + y_2)] = \frac{1}{2} \times 17 = \frac{17}{2}$.

39 [Challenging] • Target: Volumes of revolution: washer and shell methods • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Using the trapezium rule with 2 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 9, y_1 = 10, y_2 = 4$.

a. $\frac{35}{2}$

b. $\frac{23}{2}$

c. $\frac{199}{12}$

d. $\frac{33}{2}$ ✓

$\int y\,dx \approx \frac{h}{2}[y_0 + y_2 + 2(y_1 + \cdots + y_1)] = \frac{1}{2} \times 33 = \frac{33}{2}$.

40 [Challenging] • Target: A well-designed word problem: designing a vase • Cross-Lesson: 10.01 Numerical Methods

Using the trapezium rule with 2 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 5, y_1 = 7, y_2 = 8$.

a. $\frac{27}{2}$ ✓

b. $\frac{163}{12}$

c. 10

d. $\frac{29}{2}$

$\int y\,dx \approx \frac{h}{2}[y_0 + y_2 + 2(y_1 + \cdots + y_1)] = \frac{1}{2} \times 27 = \frac{27}{2}$.

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41 [Challenging] • Target: Standard Integrals (Trig, Exp & Rational) • Cross-Lesson: 11.01 Differential Equations

Using the trapezium rule with 2 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 10, y_1 = 5, y_2 = 5$.

a. 10

b. $\frac{151}{12}$

c. $\frac{27}{2}$

d. $\frac{25}{2}$ ✓

$\int y\,dx \approx \frac{h}{2}[y_0 + y_2 + 2(y_1 + \cdots + y_1)] = \frac{1}{2} \times 25 = \frac{25}{2}$.

42 [Challenging] • Target: Integration by Substitution • Cross-Lesson: 12.01 Poisson Distribution

Using the trapezium rule with 5 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 14, y_1 = 8, y_2 = 10, y_3 = 9, y_4 = 7, y_5 = 5$.

a. $\frac{89}{2}$

b. $\frac{53}{2}$

c. $\frac{87}{2}$ ✓

d. $\frac{523}{12}$

$\int y\,dx \approx \frac{h}{2}[y_0 + y_5 + 2(y_1 + \cdots + y_4)] = \frac{1}{2} \times 87 = \frac{87}{2}$.

43 [Challenging] • Target: Integration by Parts • Cross-Lesson: 13.01 Continuous Random Variables

Using the trapezium rule with 4 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 2, y_1 = 8, y_2 = 5, y_3 = 13, y_4 = 5$.

a. $\frac{33}{2}$

b. $\frac{355}{12}$

c. $\frac{61}{2}$

d. $\frac{59}{2}$ ✓

$\int y\,dx \approx \frac{h}{2}[y_0 + y_4 + 2(y_1 + \cdots + y_3)] = \frac{1}{2} \times 59 = \frac{59}{2}$.

44 [Challenging] • Target: Partial Fractions Integration & Trapezium Rule • Cross-Lesson: 14.01 Sampling & Estimation

Using the trapezium rule with 4 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 7, y_1 = 11, y_2 = 6, y_3 = 13, y_4 = 12$.

a. $\frac{81}{2}$

b. $\frac{475}{12}$

c. $\frac{79}{2}$ ✓

d. $\frac{49}{2}$

$\int y\,dx \approx \frac{h}{2}[y_0 + y_4 + 2(y_1 + \cdots + y_3)] = \frac{1}{2} \times 79 = \frac{79}{2}$.

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45 [Challenging] • Target: Integration by substitution • Cross-Lesson: 15.01 Hypothesis Testing

Using the trapezium rule with 4 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 13, y_1 = 15, y_2 = 9, y_3 = 13, y_4 = 12$.

a. $\frac{99}{2}$ ✓

b. 31

c. $\frac{595}{12}$

d. $\frac{101}{2}$

$\int y\,dx \approx \frac{h}{2}[y_0 + y_4 + 2(y_1 + \cdots + y_3)] = \frac{1}{2} \times 99 = \frac{99}{2}$.

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